

Western Ghats Field Validator

Application Demonstration Report

Executive Summary

The Western Ghats Field Validator is a mobile-first progressive web application designed to validate satellite-derived land use and land cover (LULC) classifications through ground-truth field observations. The application integrates multiple authoritative geospatial data sources including CoRE Stack APIs, Google Dynamic World, and custom forest typology datasets to provide field researchers with actionable insights at any location.

1. Application Overview

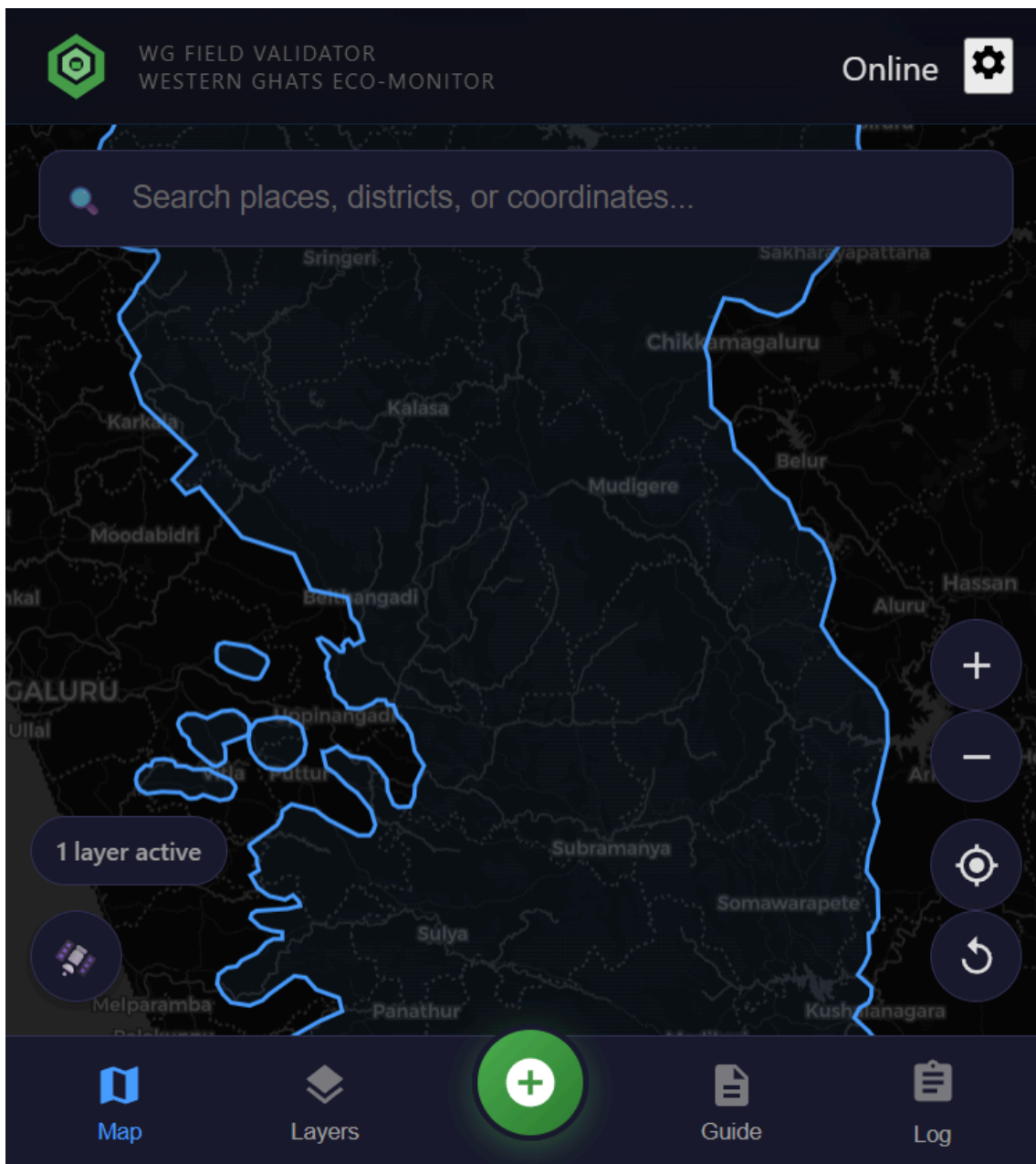


Figure 1: Main application interface showing the Western Ghats region with active layers

The application provides:

- Interactive map interface with multiple base map options (OSM, Satellite)
- GPS-based location tracking for field data collection
- Offline-first architecture for remote field work
- Integration with CoRE Stack watershed data APIs
- Dynamic World LULC classification layers
- Custom forest typology layers distinguishing old growth from plantations

2. Layer Management System

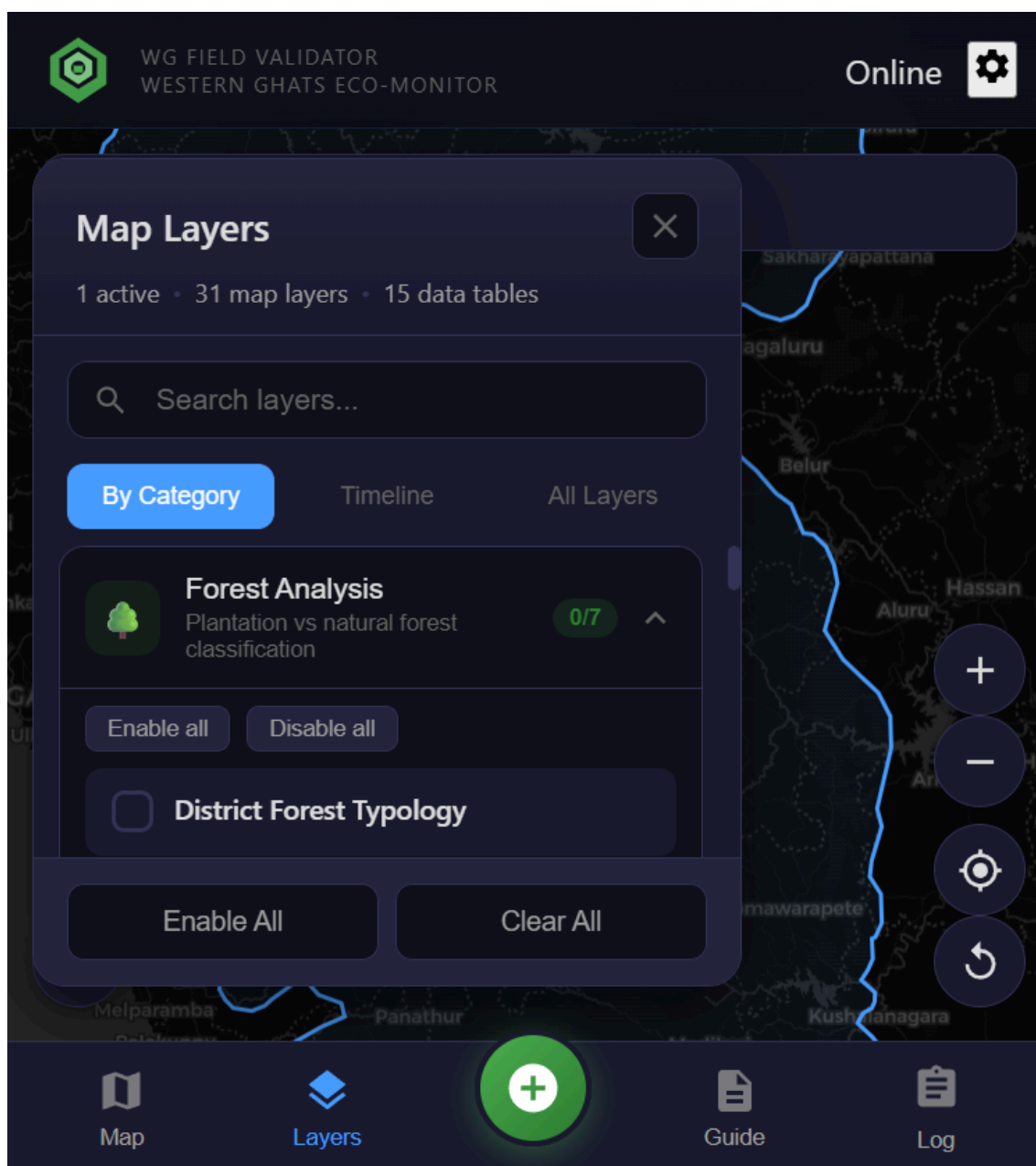


Figure 2: Layer panel showing organized categories of geospatial data

The application organizes 31 map layers and 15 data tables across six categories:

Category	Description	Layers
Forest Analysis	Plantation vs natural forest classification	7 layers
Land Cover Maps	Historical LULC from GLC-FCS30D	16 layers
Urban Expansion	Built-up area tracking over time	14 layers
Dynamic World	Live/derived LULC from Google Earth Engine	2 layers

Boundaries	Administrative boundaries	2 layers
Watershed Data	Water balance and cropping intensity	4 layers

3. Forest Classification Layers

3.1 Old Growth Forest Layer

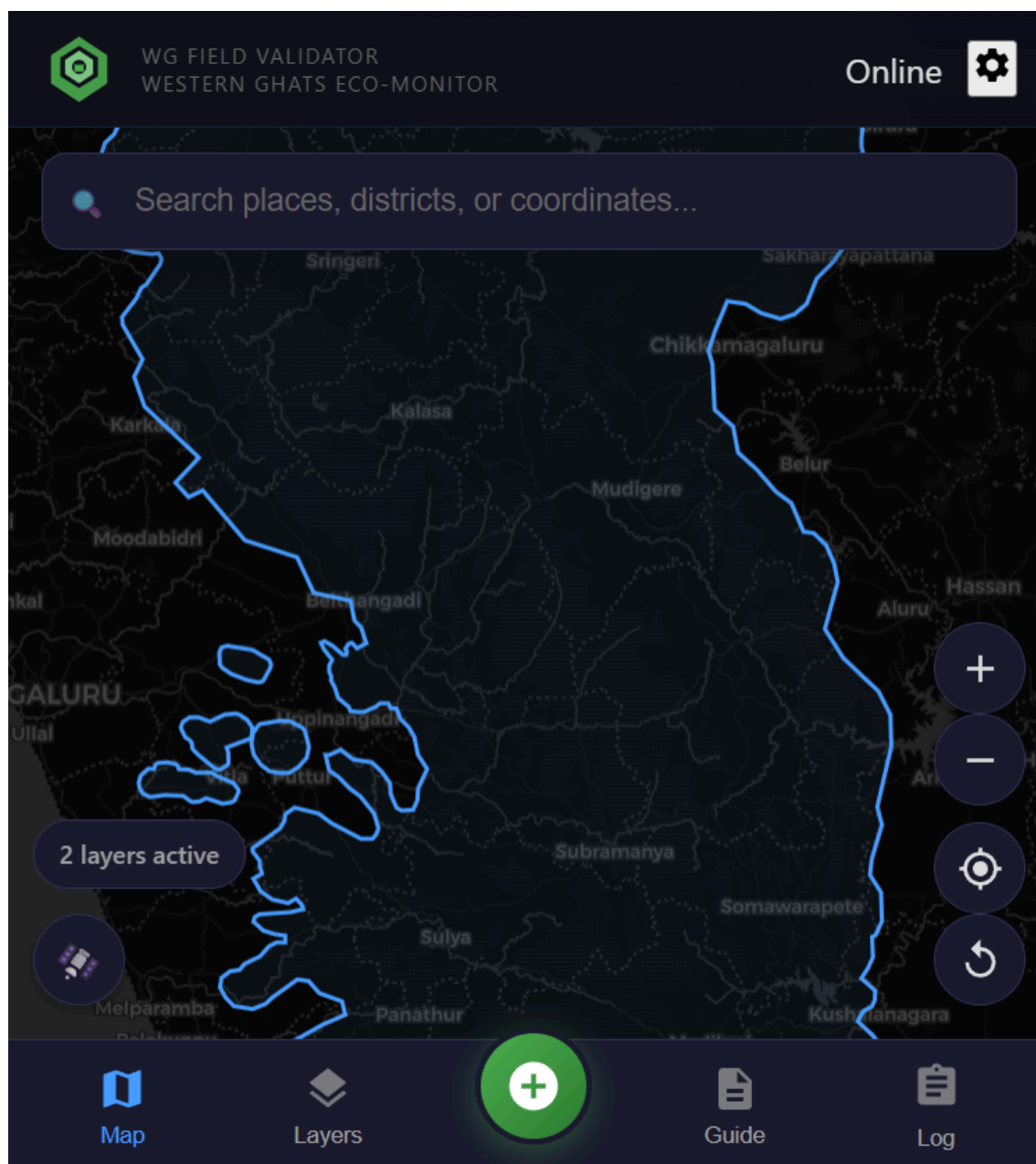


Figure 3: Old Growth Forest classification overlaid on the Western Ghats

The Old Growth Forest layer identifies mature, undisturbed forest areas using a combination of:

- Tree cover persistence (2000-2020)

- Height structure analysis
- Historical land use patterns
- Spatial connectivity metrics

3.2 Plantation Layer

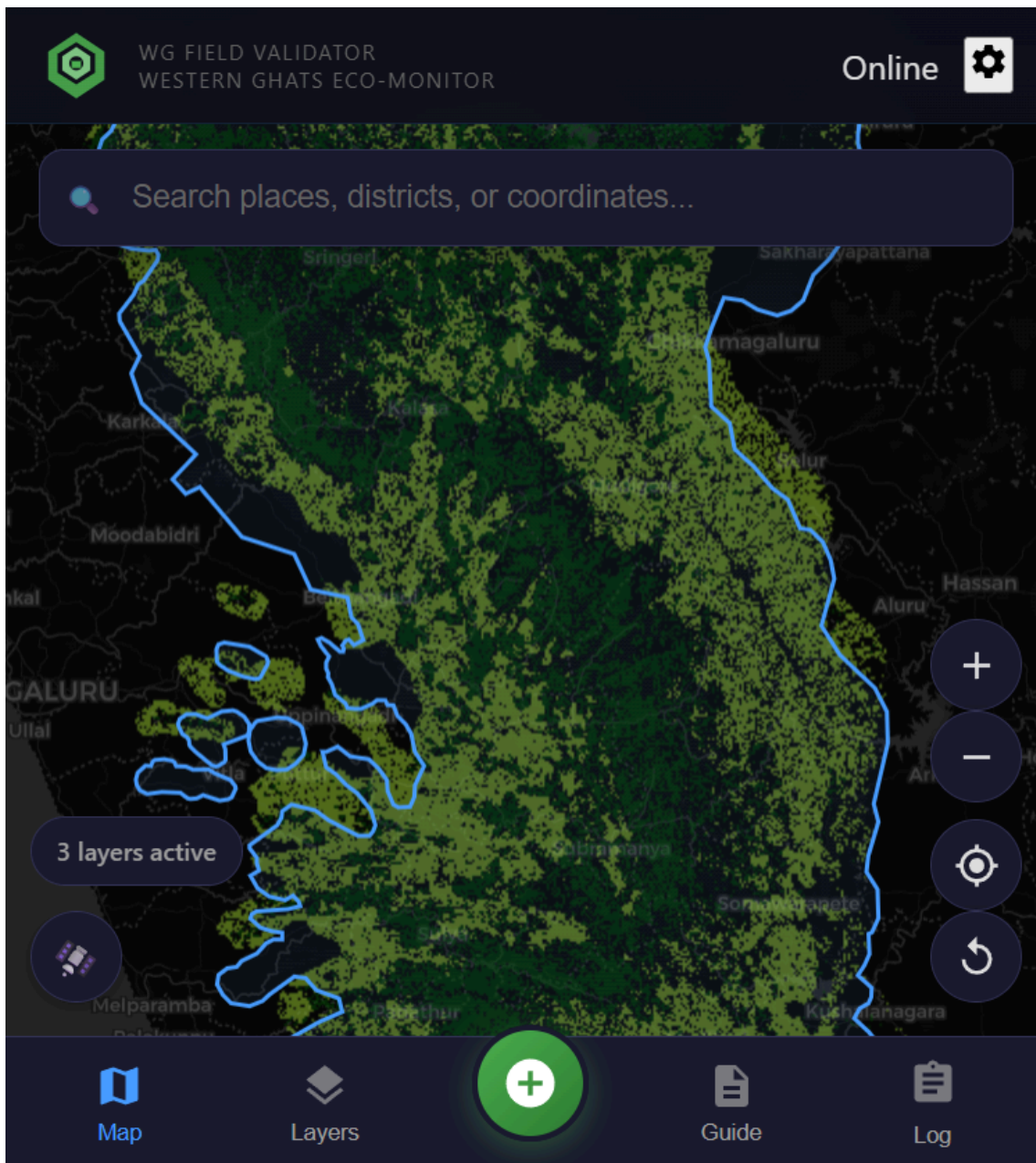


Figure 4: Combined Old Growth Forest and Plantation layers

The Plantations layer distinguishes managed tree cover (rubber, teak, eucalyptus, oil palm, etc.) from natural forests using:

- Spectral signature analysis
- Tree cover age patterns
- Row plantation detection

- Crop cycle indicators

Field Validation Use Case: Researchers can verify whether satellite-classified "forest" areas are actually old-growth natural forests or commercial plantations, a critical distinction for biodiversity assessments and conservation planning.

4. Dynamic World LULC Integration

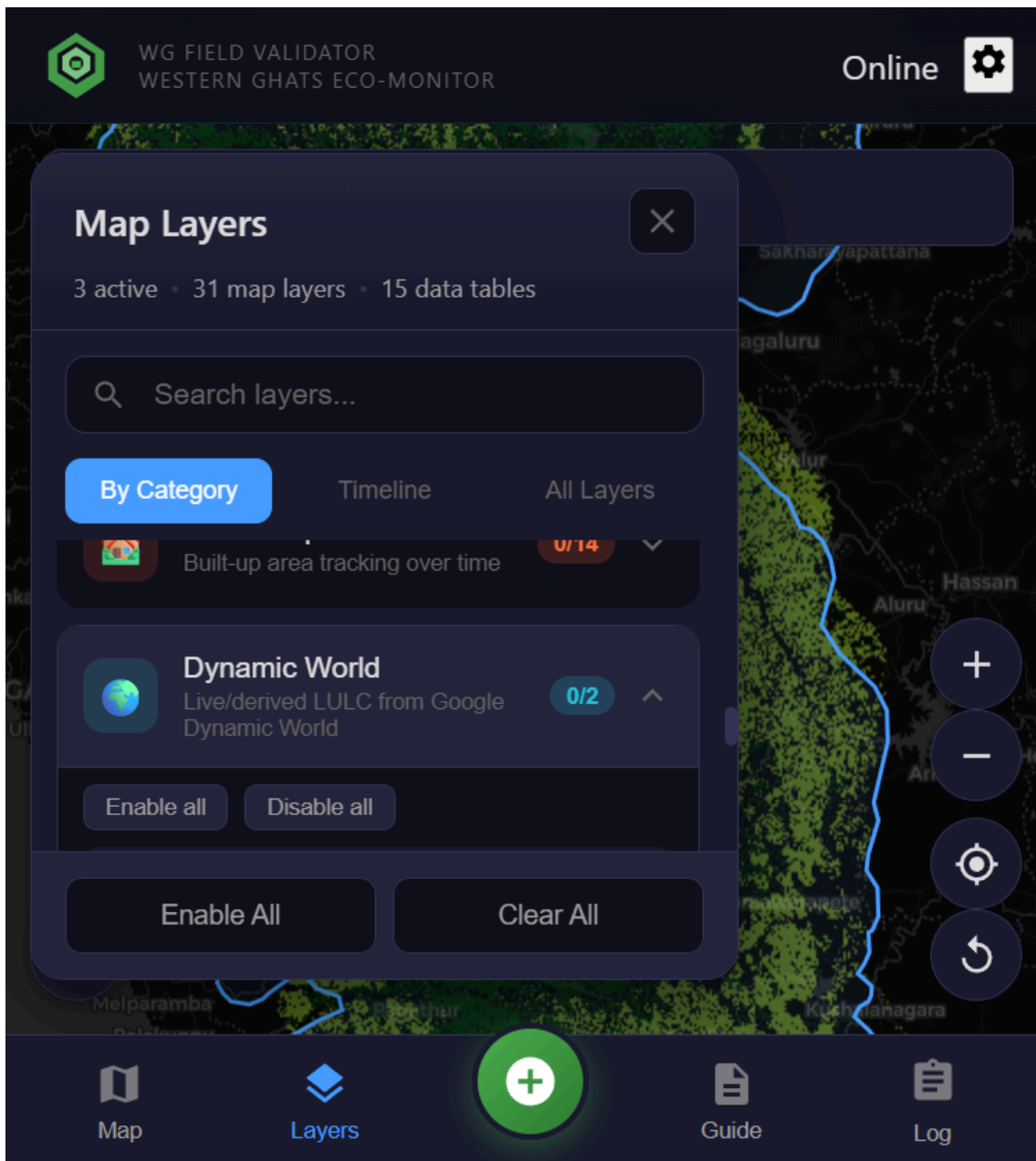


Figure 5: Dynamic World layer options including live and regional composites

The application integrates Google Dynamic World LULC data:

- **Dynamic World (Live GEE):** Near real-time land cover classification (requires API key)

- **Dynamic World Regional (2018-2025):** Pre-processed composite covering the Western Ghats

Dynamic World provides 9 land cover classes:

1. Water
 2. Trees
 3. Grass
 4. Flooded Vegetation
 5. Crops
 6. Shrub & Scrub
 7. Built-up
 8. Bare Ground
 9. Snow & Ice
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5. CoRE Stack API Integration

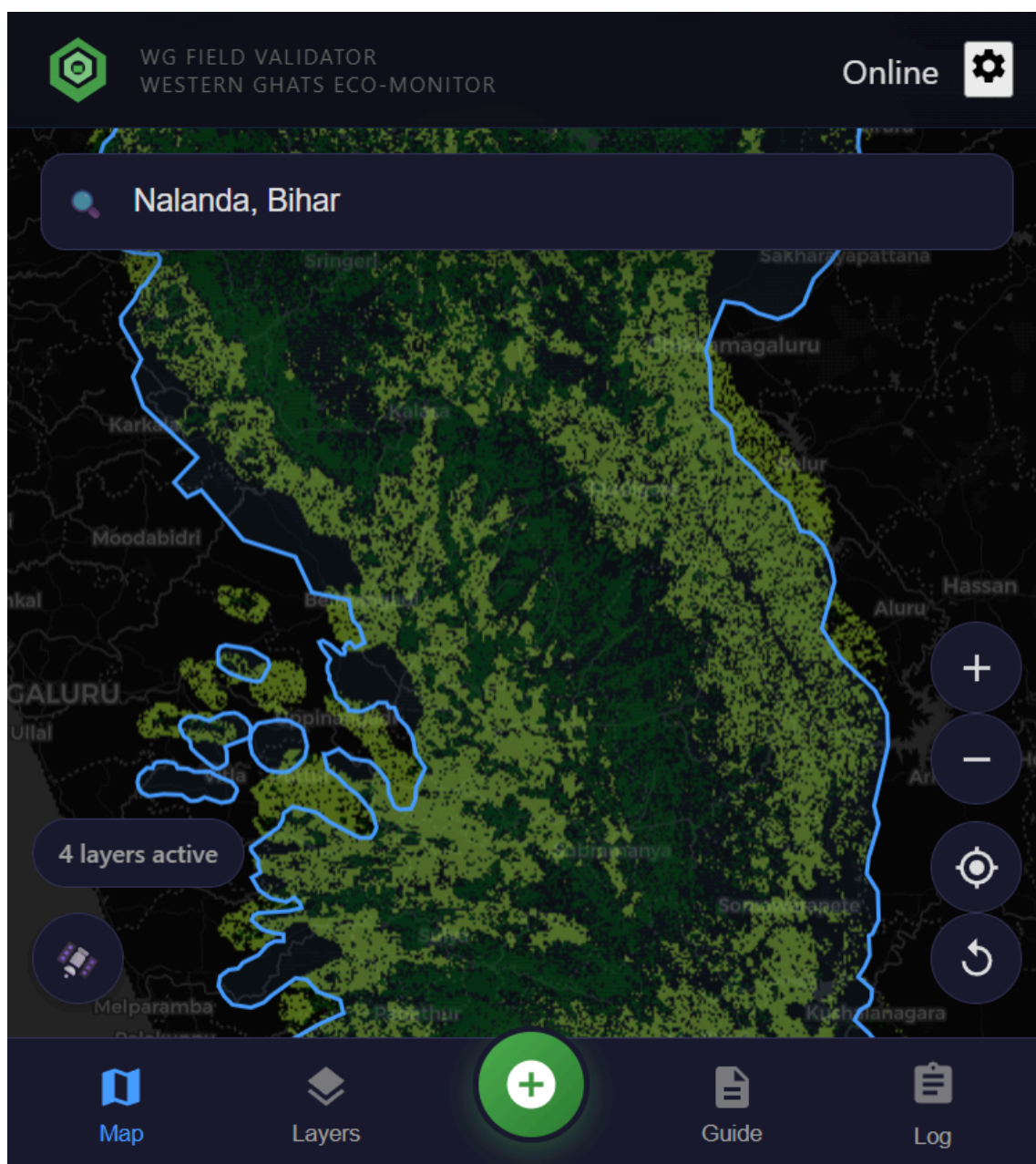


Figure 6: Map view with multiple analysis layers active

The application integrates with CoRE Stack APIs to provide:

Endpoint	Data Provided
/get_admin_details_by_latlon/	State, District, Tehsil for any coordinate
/get_mwsid_by_latlon/	Micro-Watershed ID lookup
/get_generated_layer_urls/	Available GIS layers for a tehsil
/get_tehsil_data/	Comprehensive tehsil statistics

/get_mws_data/	Time series: ET, Runoff, Precipitation
/get_mws_kyl_indicators/	Know Your Landscape indicators
/get_waterbodies_data_by_admin/	Waterbody inventory
/get_mws_report/	MWS assessment report URL

Data Coverage: Bihar and Uttar Pradesh have comprehensive CoRE Stack data coverage, enabling watershed-level analysis for field validation in these regions.

6. Field Guide and Protocols

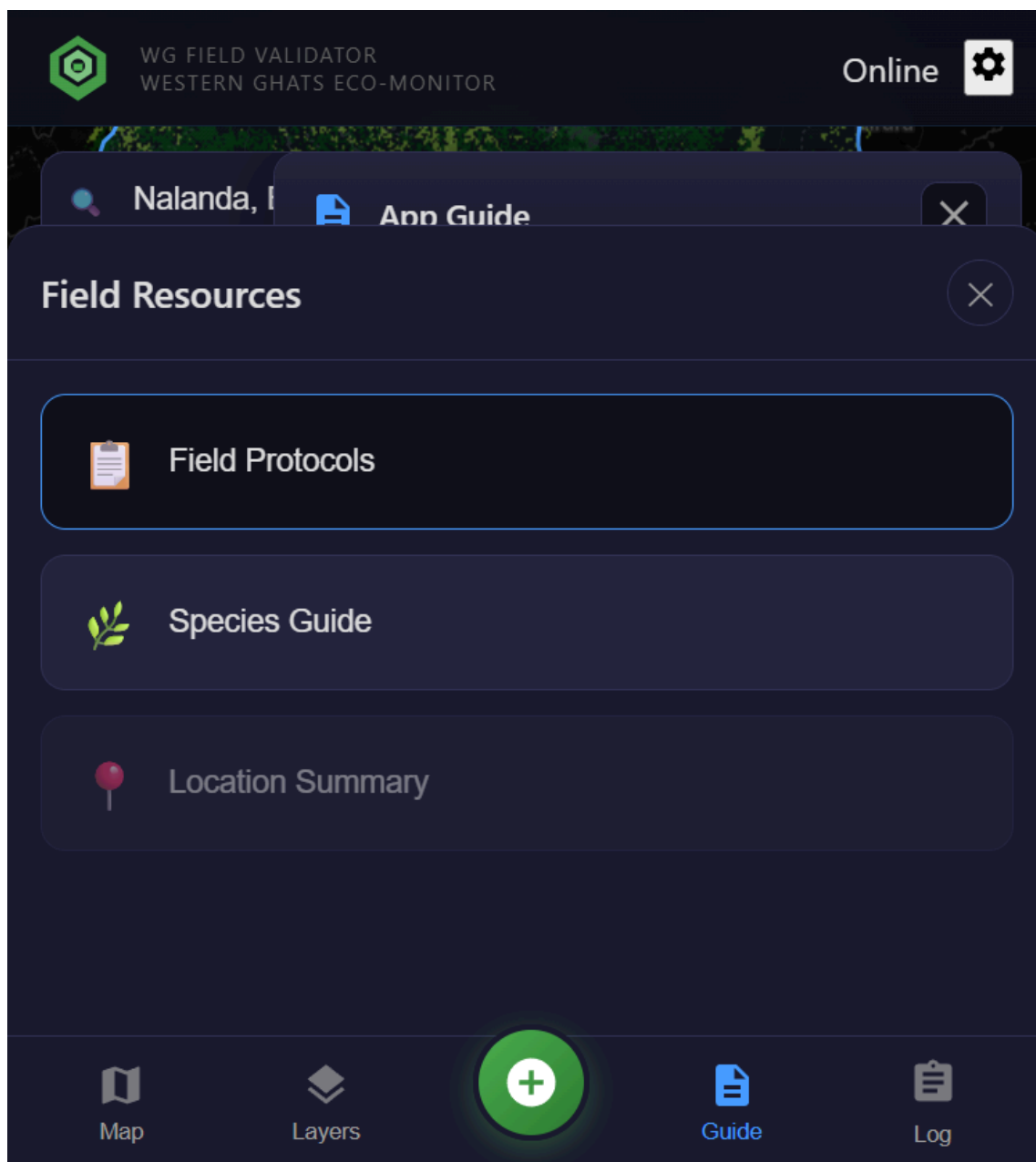


Figure 7: Built-in field guide with usage instructions

The application includes comprehensive field guides covering:

- Getting started and GPS configuration
- Navigation and map controls
- Layer management
- Observation capture workflow
- Location information panel
- Field log management
- Offline mode operation

7. Field Observation Capture

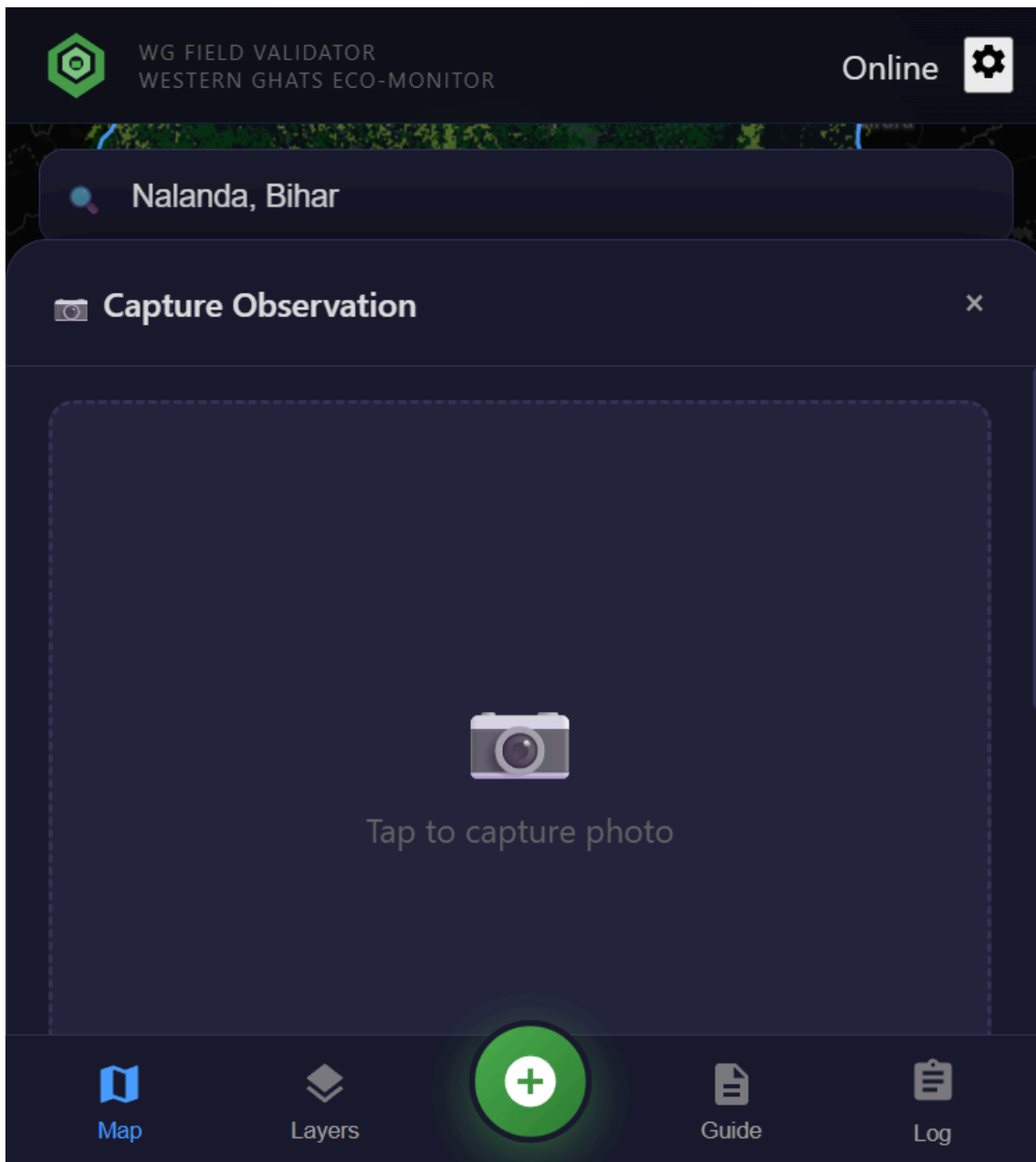


Figure 8: Field observation capture interface

The observation capture workflow enables:

1. **Photo Documentation:** Camera or gallery image capture
2. **Automatic Location:** GPS coordinates with accuracy indicator
3. **Dataset Values:** Active layer values at current location
4. **Validation Status:**
 - Match: Satellite classification matches ground truth
 - Mismatch: Classification differs from observed land cover
 - Unclear: Unable to determine classification accuracy

8. Field Log and Data Export

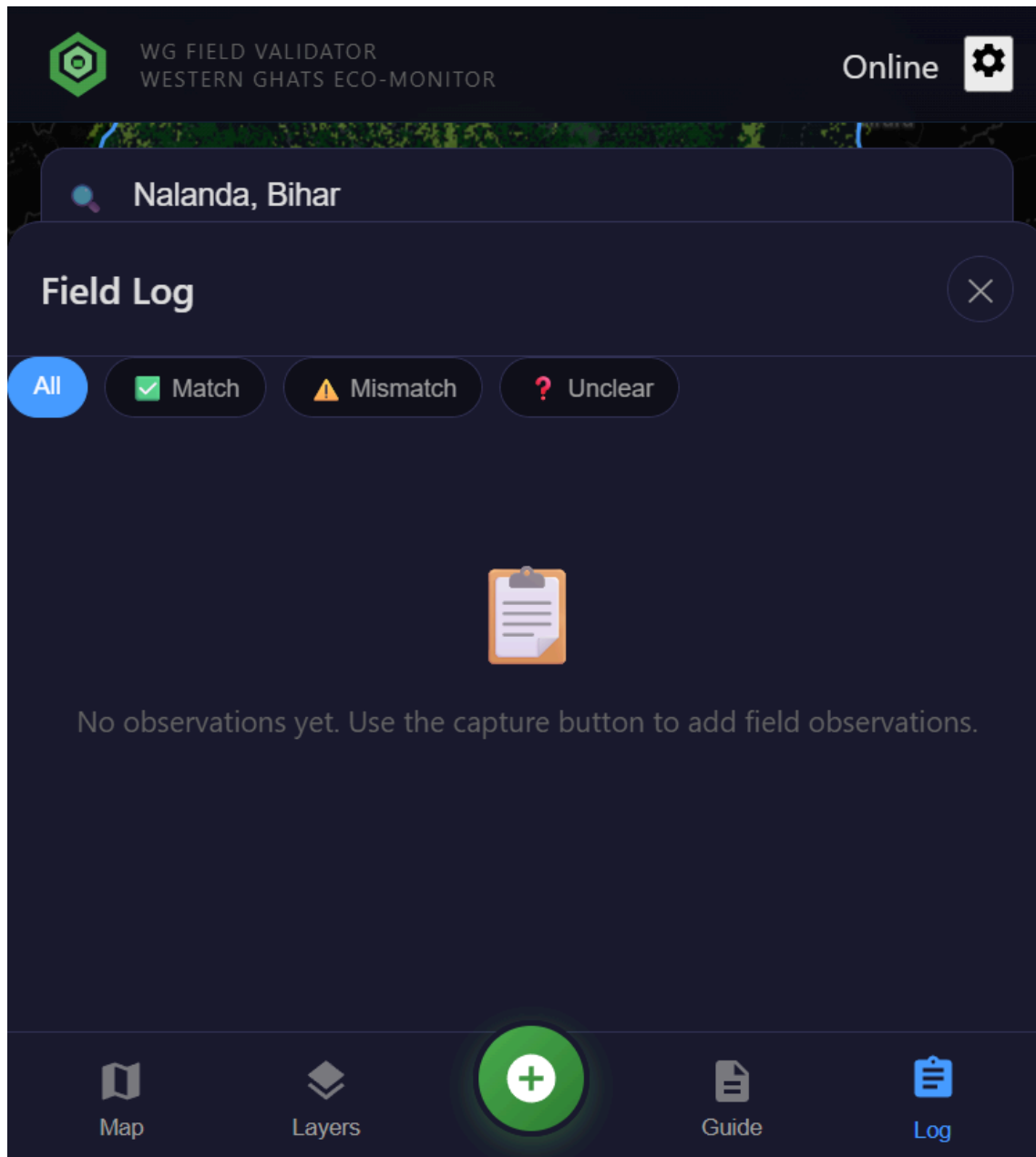


Figure 9: Field log showing captured observations with filtering

The Field Log provides:

- Chronological list of all field observations
 - Filtering by validation status (Match/Mismatch/Unclear)
 - Export to JSON and CSV formats
 - Offline storage with sync on connectivity
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9. Satellite Base Map

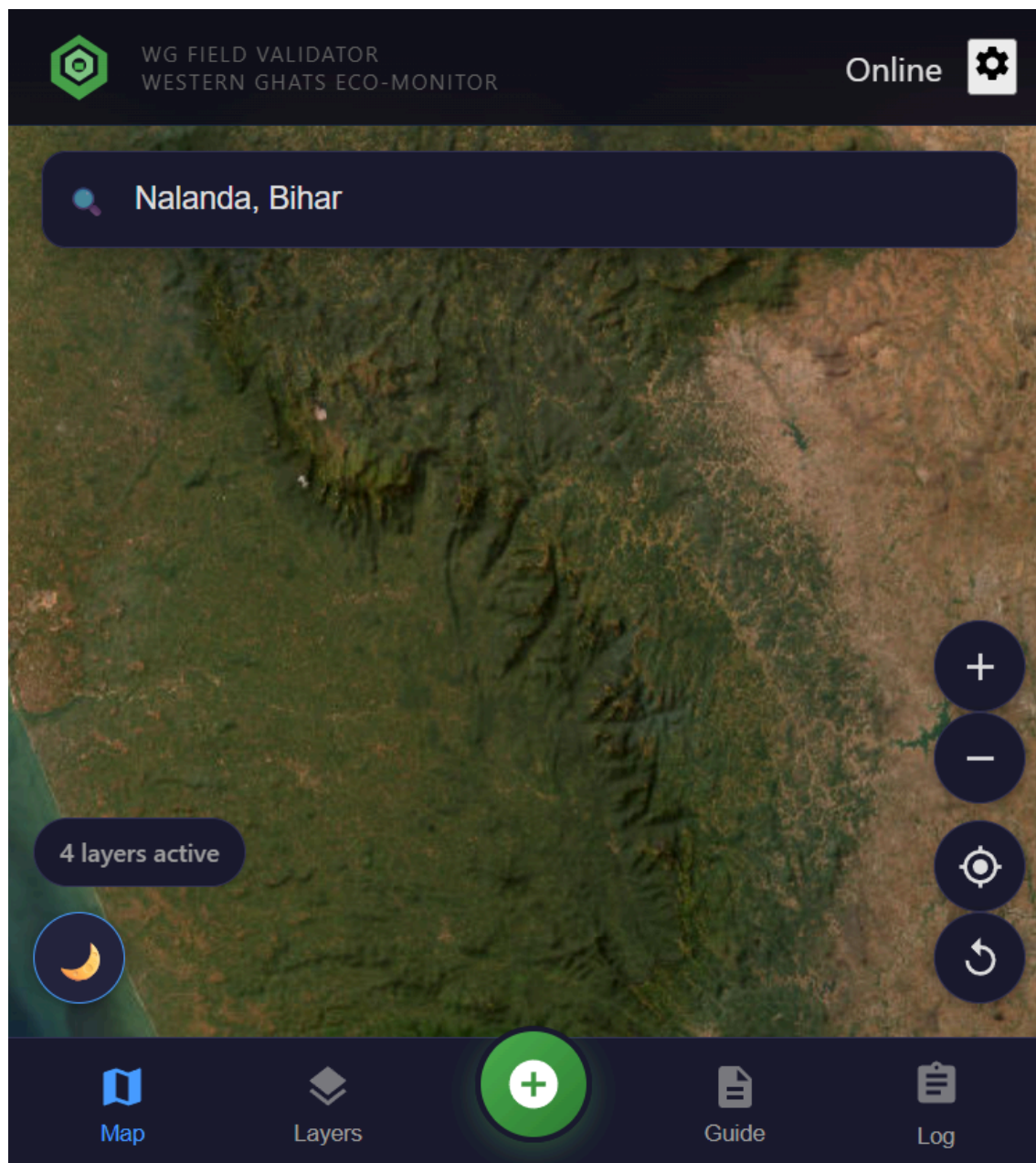


Figure 10: Satellite imagery base map for visual verification

The application supports multiple base maps:

- OpenStreetMap (default)
- ESRI Satellite Imagery
- Dark mode for low-light conditions

10. Key Benefits

For Field Researchers

- **Offline Capability:** Works without internet connectivity in remote areas
- **GPS Integration:** Automatic location capture with accuracy indicators
- **Multi-Layer Analysis:** Compare multiple data sources simultaneously
- **Photo Documentation:** Visual evidence linked to coordinates

For Conservation Organizations

- **Ground Truth Collection:** Systematic validation of satellite classifications
- **Plantation vs Forest Distinction:** Critical for biodiversity assessments
- **Historical Analysis:** Track land cover changes over time
- **Standardized Protocols:** Consistent data collection methodology

For Data Scientists

- **API Integration:** Tested CoRE Stack API endpoints
- **Export Formats:** JSON and CSV for analysis workflows
- **Reproducibility:** Documented methodology and sample data

11. Technical Architecture

Component	Technology
Frontend	React 18 + TypeScript + Vite
Mapping	MapLibre GL JS
Mobile	Capacitor (Android APK)
Offline Storage	IndexedDB
APIs	CoRE Stack, OpenStreetMap Nominatim
Deployment	Static hosting (GitHub Pages compatible)

12. Sample Field Data

Sample field observations are available at: `field-data/recovered-observations/`

This includes real validation data collected in the Western Ghats region.

13. Getting Started

1. Clone the repository
2. Copy `.env.example` to `.env` and configure API keys
3. Run `npm install` followed by `npm run dev`
4. For Android: `npm run build && npx cap sync android`

Appendix: API Testing Notebook

A Jupyter notebook demonstrating all tested CoRE Stack API endpoints is available at:
`docs/notebooks/corestack_api_tested.ipynb`

